GIDEON STEIN

Machine Learning Researcher

gideon.stein@uni-jena.de | Gideon-Stein | +49 176 4565 1723 | Humboldtstraße 19, 07743 Jena
gideon-stein.github.io/

EDUCATION

PhD: Causal Discovery in Real-world Applications

FSU Jena | May 2021 – Present | Jena, Germany

- PhD project in cooperation with the CVG Jena and iDiv
- Focus on the construction of large-scale benchmarks for causal discovery with downstream applications in biodiversity research

M.Sc. Machine Learning & Data Analysis

ITMO University | 2018 – 2020 | St. Petersburg, Russia

- Independent research publications on Reinforcement Learning
- Awarded "Best Thesis" by ITMO University

B.A. Philosophy & Economics

University of Bayreuth | 2014 – 2018 | Bayreuth, Germany

- Focus on Game Theory and Decision Theory

SELECTED PUBLICATIONS

G. Stein, et al. "TCD-Arena: Assessing rrobustness of time series causal discovery against assumption violations". **ICLR 2026.**

G. Stein, et al. "CausalRivers – Scaling up benchmarking of causal discovery for real-world time-series". **ICLR Spotlight 2025.**

Yuanyuan Huang* & G. Stein*, et al. "Enhanced stability of grassland soil temperature by plant diversity". **Nature Geoscience 2024.**

G. Stein, et al. "Embracing the black box: Heading towards foundation models for causal discovery from time series". **AAAI WS 2024.**

INDUSTRY AND RESEARCH

Research Associate: KI4KI - Remote Sensing for Large Infrastructure monitoring

FSU Jena | May 2022 – Present | Jena, Germany

- Research on satellite-based dam deformation forecasting

Freelance ML Consultant & Software Engineering

steinML / EasyMedBox / bliro.io / Desko | Sep 2020 – Present | Remote / Jena, Germany

- Machine learning consulting
- Feasibility studies on VLMs and ASR systems.
- Early-stage product development for tech startups (EasyMedBox, Bliro) parallel to doctoral research.

ABOUT ME

I am a Machine Learning Researcher and PhD candidate specializing in **Causal Discovery, Time Series Processing, and Deep Learning.** My vision is to build automated systems that extract causal structure from observational data, fundamentally rethinking how we uncover complex relationships in both the sciences and the private sector. Currently I am laying the groundwork for this by constructing large-scale benchmarks and working on foundation models.

SKILLS

Research Focus

Causal Discovery | Time-Series Processing | Deep Learning | Benchmarking | Reinforcement Learning

Frameworks and Tools

Python | PyTorch | PyTorch Lightning | Slurm | Docker | Hydra | Git | LaTeX | Open-CV | Scikit | Pandas | NumPy

LANGUAGES

English ●●●●●

German ●●●●●

REFEREES

Prof. Dr. Joachim Denzler
FSU Jena
joachim.denzler@uni-jena.de

Prof. Dr. Nico Eisenhauer
iDiv
nico.eisenhauer@idiv.de

Assoc. Prof. Andrey Filchenkov
ITMO University
afilchenkov@itmo.ru